

Databricks Interview Questions & Answers

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databricks

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- 2)What are the types of Databricks Runtimes?
- 3)How to share Notebook to other Developers in Workspace?
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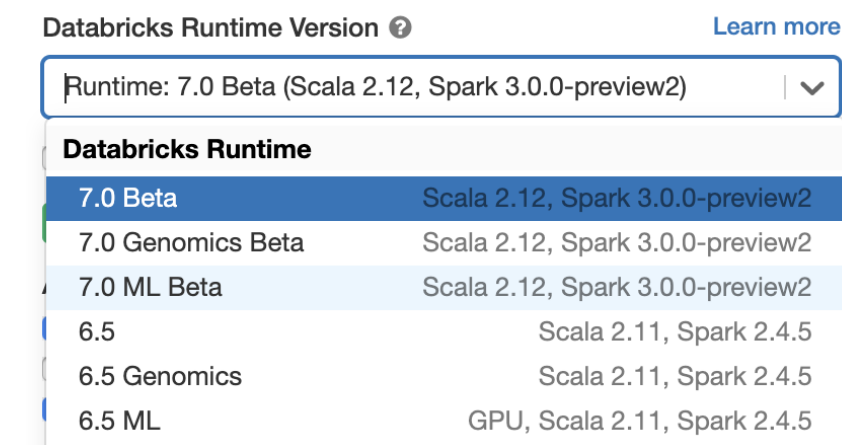
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1) What is Databricks Runtime?

The set of core components that run on the clusters managed by Databricks. Consists of the underlying Ubuntu OS, pre-installed languages and libraries (Java, Scala, Python, and R), Apache Spark, and various proprietary Databricks modules (e.g. DBIO, Databricks Serverless, etc.).

Azure Databricks offers several types of runtimes and several versions of those runtime types in the Databricks Runtime Version drop-down when you create or edit a cluster.



2) What are the types of Databricks Runtimes?

There are major 4 types of Databricks Runtimes.

- Databricks Runtime for Standard
- Databricks Runtime for Machine Learning
- Databricks Runtime for Genomics
- Databricks Light

Databricks Runtime for Standard

Databricks Runtime includes Apache Spark but also adds a number of components and updates that substantially improve the usability, performance, and security of big data analytics.

Databricks Runtime for Machine Learning

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Databricks Runtime ML is a variant of Databricks Runtime that adds multiple popular machine learning libraries, including TensorFlow, Keras, PyTorch, and XGBoost. ML also supports additional GPU supporting libraries clusters. **Graphics processing Units** Speeding up Machine Learning models. GPUs can drastically lower the cost because they support efficient parallel computation.

Databricks Runtime for Genomics

Databricks Runtime for Genomics is a variant of Databricks Runtime optimized for working with genomic and biomedical data.

Databricks Light

Databricks Light provides a runtime option for jobs that don't need the advanced performance, reliability, or autoscaling benefits provided by Databricks Runtime.



Databricks Light does not support:

- Delta Lake
- Autopilot features such as autoscaling
- Highly concurrent, all-purpose clusters
- Notebooks, dashboards, and collaboration features
- Connectors to various data sources and BI tools

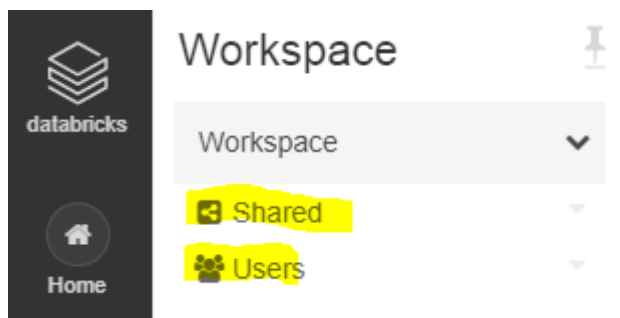
3) How to share Notebook to other Developers in Workspace?

There are two ways we can share notebooks to another developers.

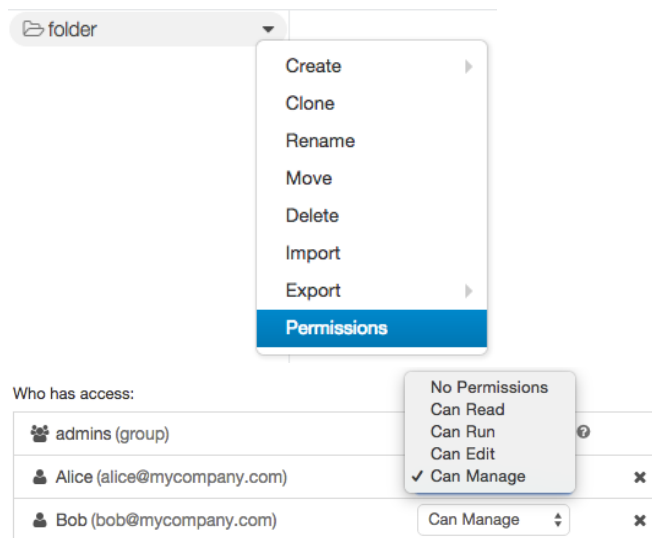
- a) Copying notebook into from user folder to shared folder.

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b) Or Giving Access to other developers from current folder or user folder.



4) How to access one notebook variable into other notebooks?

If we run a one notebook into another notebook using `%run` we can use all functions, variables and imported libraries from callee notebook to caller notebook.

```
Python
%run /Users/path/to/notebookA
```

5) How to call one notebook from another notebook?

There are two ways we can call one notebook into another notebook.

1. `%run notebook_name`

```
Python
%run /Users/path/to/notebookA
```

If we use the `%run` command. It will return from **callee notebooks** containing function and variable definitions. We can use those functions and variables in In **caller notebook**.

2. `Dbutils.notebook.run(notebook_name, timeout_sec, arguments_values)`

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You cannot use functions and variables. Only return value using arguments parameter.

```
returned_table = dbutils.notebook.run("notebook_exit_2", 60)
display(sqlContext.read.parquet(returned_table))
```

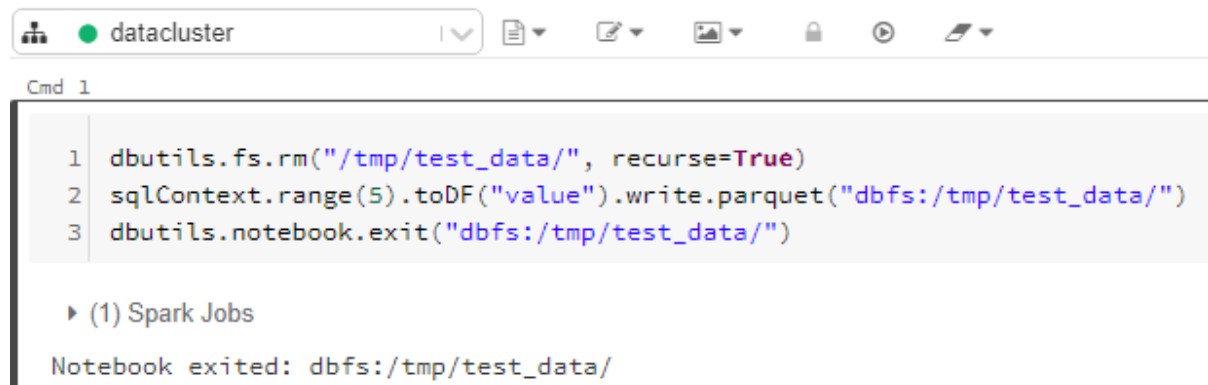
6) How to exit a notebook with returning some output data?

dbutils.notebook.exit (value: String): void -> This method lets you exit a notebook with a value.

In this example we have two notebooks. 1 for running exit and passing input value and another notebook for running 1st notebook using `dbutils.notebook.run()` method. And storing into variable.

1st notebook

notebook_exit_2 (Python)



2nd notebook.

```
returned_table = dbutils.notebook.run("notebook_exit_2", 60)
display(sqlContext.read.parquet(returned_table))
```

7) How to create Internal & External tables in Databricks?

A Databricks database is a collection of tables. A Databricks table is a collection of structured data. You can cache, filter, and perform any operations supported by Apache Spark DataFrames on Databricks tables. You can query tables with Spark APIs and Spark SQL.

External Table.

The table uses the custom directory specified with `LOCATION`. Queries on the table access existing data previously stored in the directory. When an `EXTERNAL` table is dropped, its data is not deleted from the file system. This flag is implied if `LOCATION` is specified.

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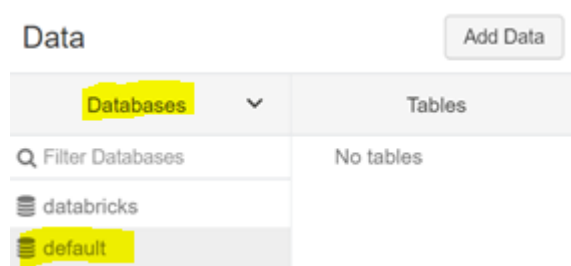
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```
CREATE EXTERNAL TABLE IF NOT EXISTS my_table (name STRING, age INT)
COMMENT 'This table is created with existing data'
LOCATION 'spark-warehouse/tables/my_existing_table'
```

Internal or Managed Table

Create a managed table using the definition/metadata of an existing table or view. The created table always uses its own directory in the default warehouse location.

```
CREATE TABLE boxes
(width INT, length INT, height INT)
USING PARQUET
OPTIONS ('compression'='snappy')
```



8) How to Access ADLS or Blob Storage in Databricks?

We can Mount Azure Blob storage containers to DBFS and we can access through DBFS mount point.

You can mount a Blob storage container or a folder inside a container to Databricks File System (DBFS) using `dbutils.fs.mount`. The mount is a pointer to a Blob storage container, so the data is never synced locally.

```
dbutils.fs.mount(
  source = "wasbs://<container-name>@<storage-account-name>.blob.core.windows.net",
  mount_point = "/mnt/<mount-name>",
  extra_configs = {"<conf-key>":dbutils.secrets.get(scope = "<scope-name>", key = "<key-name>")})
```

Access files in your container as if they were local files, for example:

```
# python
df = spark.read.text("/mnt/<mount-name>/...")
df = spark.read.text("dbfs:/<mount-name>/...")
```

Once an account access key or a SAS is set up in your notebook, you can use standard Spark and Databricks APIs to read from the storage account

Set up an account access key:

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```
spark.conf.set(
  "fs.azure.account.key.<storage-account-name>.blob.core.windows.net",
  "<storage-account-access-key>")
```

Set up a SAS for a container:

```
spark.conf.set(
  "fs.azure.sas.<container-name>.<storage-account-name>.blob.core.windows.net",
  "<complete-query-string-of-sas-for-the-container>")
```

9) What are the types of Cluster Modes in Databricks?

- a) Standard clusters
- b) High concurrency clusters

10) What are the types of workloads we can use in Standard type Cluster?

There are three types of workloads we can use in Standard type cluster. Those are

- a. DATA ANALYTICS
- b. DATA ENGINEERING
- c. DATA ENGINEERING LIGHT

FEATURE	DATA ANALYTICS	DATA ENGINEERING	DATA ENGINEERING LIGHT
Apache Spark on Databricks platform			
Job scheduling with libraries			
Job scheduling with Notebooks			
Autopilot clusters			
Databricks Runtime for ML			
MLflow on Databricks Preview			
Databricks Delta			
Interactive clusters			
Notebooks and collaboration			
Ecosystem integrations			

11) Can I use both Python 2 and Python 3 notebooks on the same cluster?

No. In Single Cluster you can use only one Python2 or Python 3. You cannot use both Python2 and python3 on same databricks cluster.

12) What is pool? Why we use pool? How to create pool in Databricks?

Pool is used to reduce cluster start time while auto scaling, you can attach a cluster to a predefined **pool of idle instances**. When attached to a pool, a cluster allocates its driver and worker nodes from the pool. If the pool does not have sufficient idle resources to accommodate the cluster's request, the pool expands by allocating new instances from the

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instance provider. When an attached cluster is terminated, the instances it used are returned to the pool and can be reused by a different cluster.

Clusters



Clusters



Clusters Pools

+ Create Pool

Refresh

Name	Instance Type	Min Idle	Max Capacity	Idle Instances	Used Instances	Actions
Demo Pool	30.5 GB Memory, 4 Cores	2	10	2	2	

1 - 1 of 1 < > 20 / Page Go to 1

Clusters / Pools / Pool Details



Demo Pool

Edit

Delete

Refresh

Overview Configuration

Instance Type: , 30.5 GB Memory, 4 Cores

Min Idle: 2

Total instances: 2

Idle Instance Auto Termination: 60 minutes

Max Capacity: 10

Used: 0 Idle: 0 (Pending: 2) Max Capacity: 10

13) How many ways we can create variables in Databricks?

There are different ways we can create variables in Databricks.

One method is creating variable and assigning values and calling that notebook into another notebook using **%run**

```
Python
%run /Users/path/to/notebookA
```

Another method is using `dbutils.widgets` method.

Use `dbutils.widgets.text()` or `dbutils.widgets.dropdown()` to create a widget parameter or variable

and `dbutils.widgets.get()` to get its bound value.

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Var_v:

Cmd 1

```
1 dbutils.widgets.text("Var_v", "text_Value")
2 print(dbutils.widgets.get("Var_v"))
```

text_Value

Command took 0.03 seconds -- by pysparktelugu@gmail.com at 9/6/2020, 2:5

V_X:

Cmd 1

```
1 dbutils.widgets.dropdown("v_x", "1", [str(x) for x in range(1, 10)])
2 print(dbutils.widgets.get("v_x"))
```

1

If we type `.help()` it will show available methods in `dbutils.widgets`

Like **creating** text, **dropdown**, **combobox** variables and getting values using GET method.

Removing all variables using **`dbutils.widgets.removeAll()`**

Cmd 1

```
1 dbutils.widgets.help()
```

dbutils.widgets provides utilities for working with notebook widgets. You can create diff **`dbutils.widgets.help("methodName")`**.

combobox(name: String, defaultValue: String, choices: Seq, label: String): void ->
dropdown(name: String, defaultValue: String, choices: Seq, label: String): void ->
get(name: String): String -> Retrieves current value of an input widget
getArgument(name: String, optional: String): String -> (DEPRECATED) Equivalent to
multiselect(name: String, defaultValue: String, choices: Seq, label: String): void ->
remove(name: String): void -> Removes an input widget from the notebook
removeAll: void -> Removes all widgets in the notebook
text(name: String, defaultValue: String, label: String): void -> Creates a text input w

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14) What are the limitations in Jobs?

- A. The number of jobs is limited to 1000.
- B. A workspace is limited to 150 concurrent (running) job runs.
- C. A workspace is limited to 1000 active (running and pending) job runs.

15) Can I use %pip in notebook for installing packages or libraries?

Yes. We can use.

Creating a file for list of commands to be executed

```
dbutils.fs.put("/dbfs:/home/myScripts/fast.ai", "conda install -c pytorch -c fastai fastai -y", True)
```

Installing using created file

```
%pip install -r /dbfs/home/myScripts/fast.ai
```

Installing using %pip

```
%pip install matplotlib
```

Uninstalling using %pip

```
%pip uninstall -y matplotlib
```

We can use conda also same as like %pip

```
%conda install matplotlib
```

Import the file to another notebook using Conda env update.

```
%conda env update -f /dbfs/myenv.yml
```

List the Python environment of a notebook

```
%conda list
```